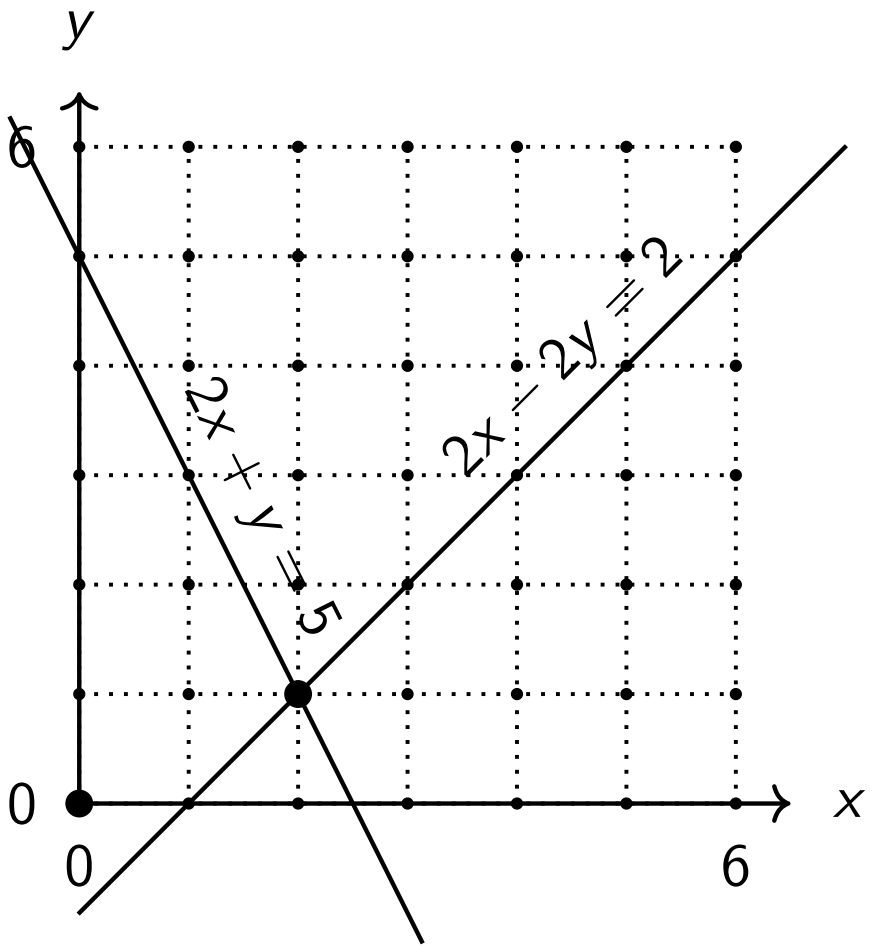
Introduction to Modern Math: Graph Theory

April 15, 2019

What is a graph?

What isn't a graph?



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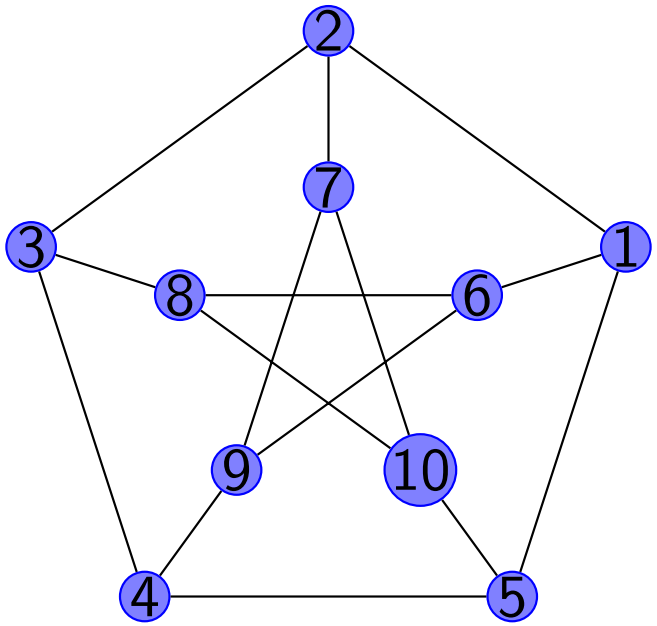

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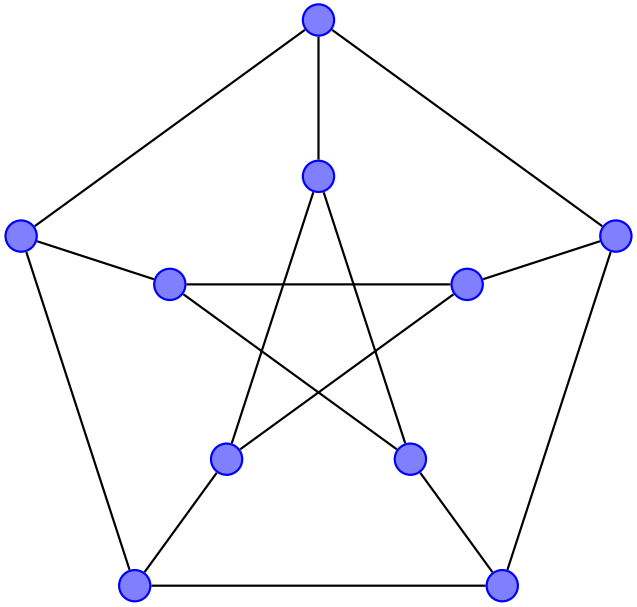

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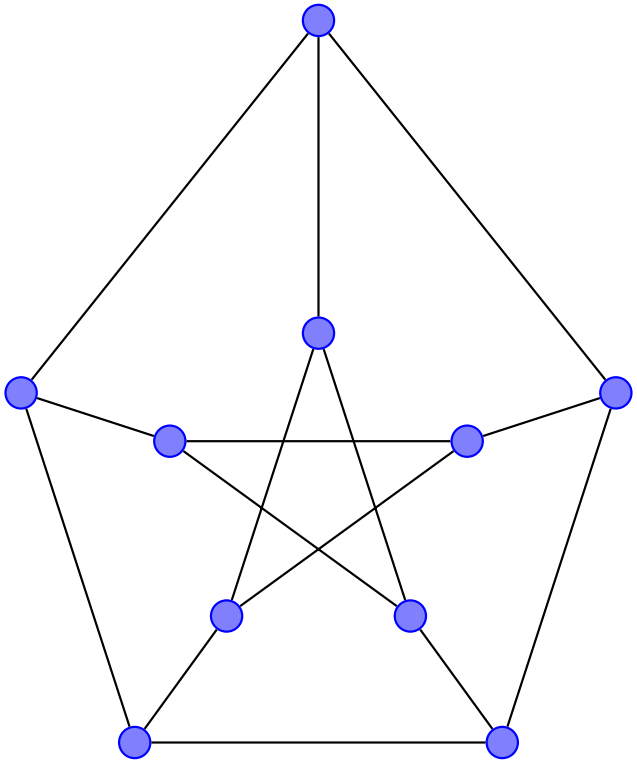

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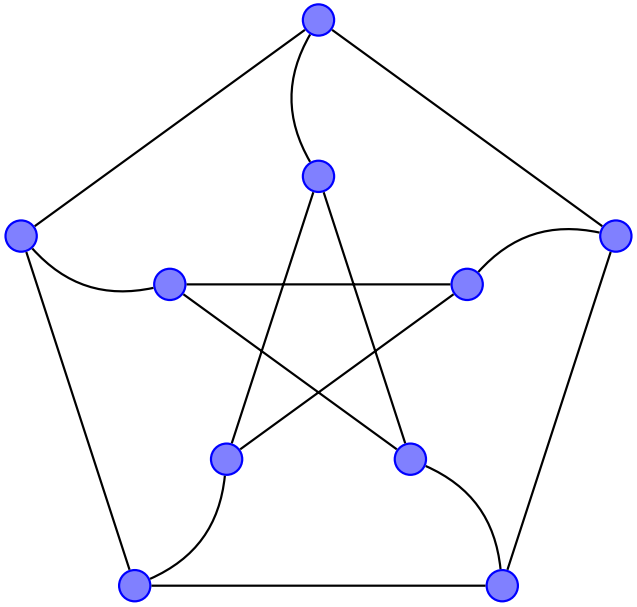

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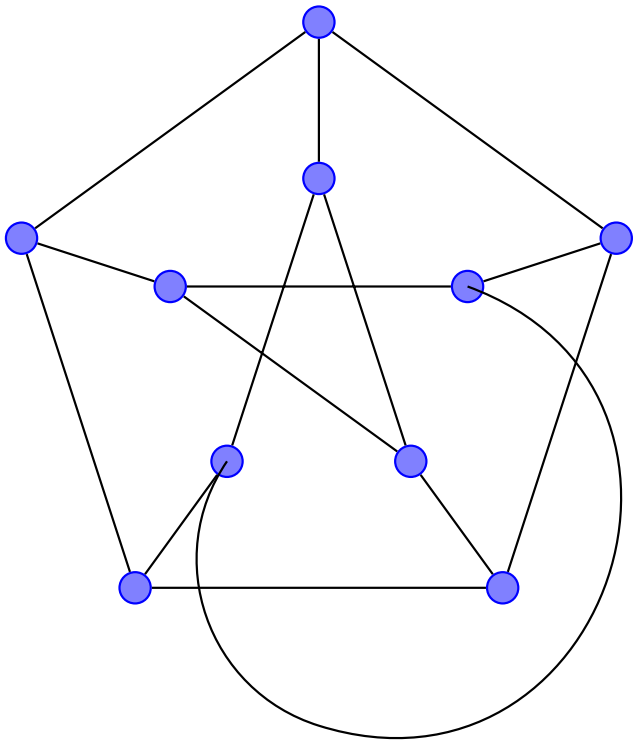
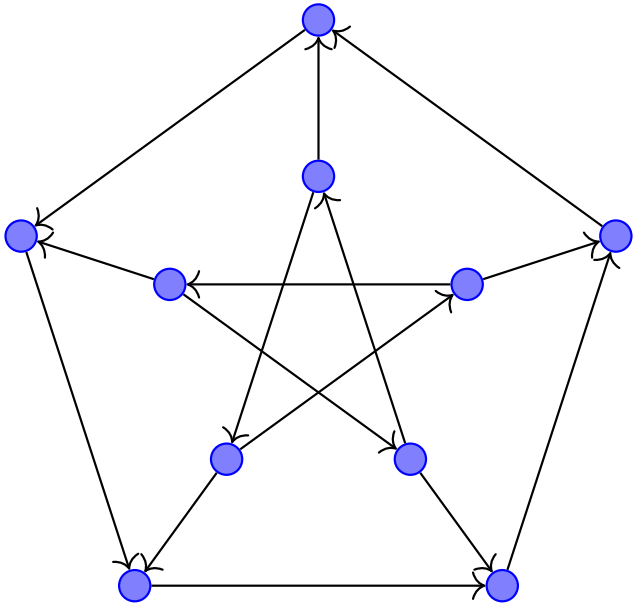


What is kind of like a graph?



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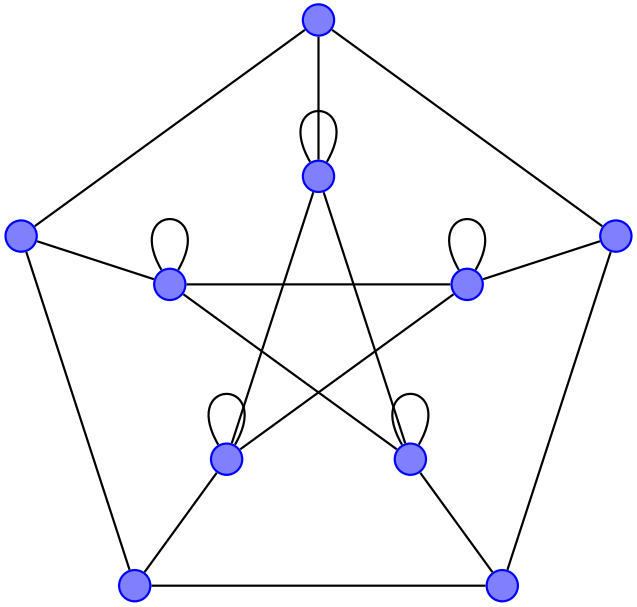

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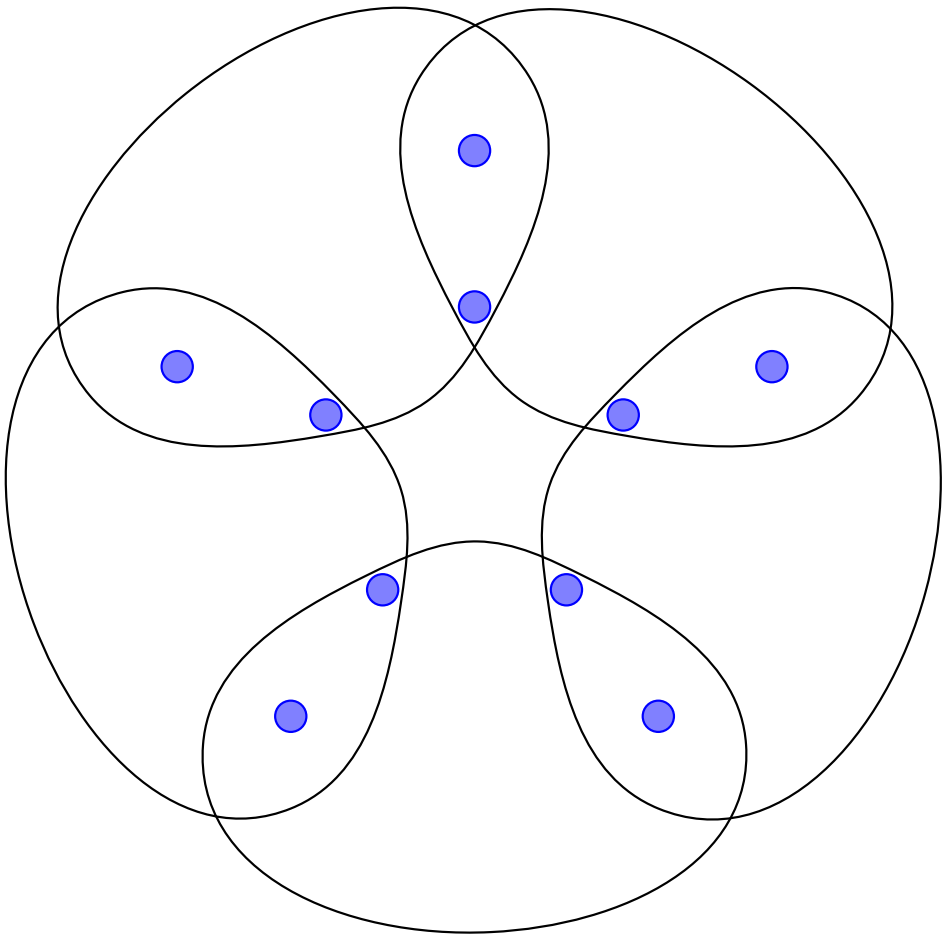


What are graphs for?

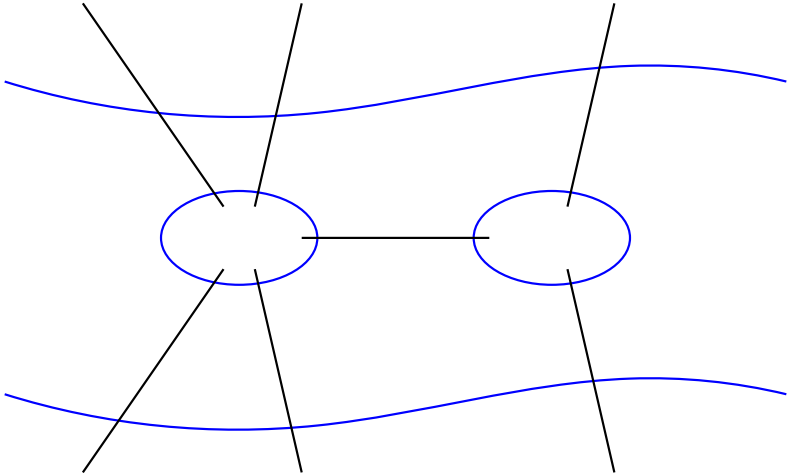
What are graphs for?

- Computers: networks, organising data and its flow
- Biology: speciation, genetics
- Chemisty: cellular structure and bonds
- Physics: electrical networks
- Sociology: social dynamics, trending
- Math: everything (adds the first level of structure above just sets)

Classical Applications and Results

Eulerian circuits and the Bridges of Konigsburg

The bridges of Konigsburg



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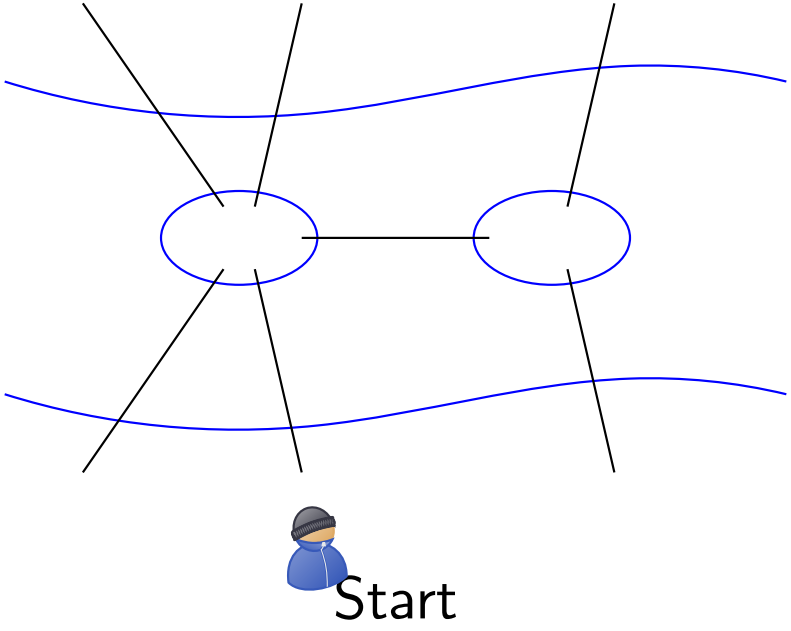

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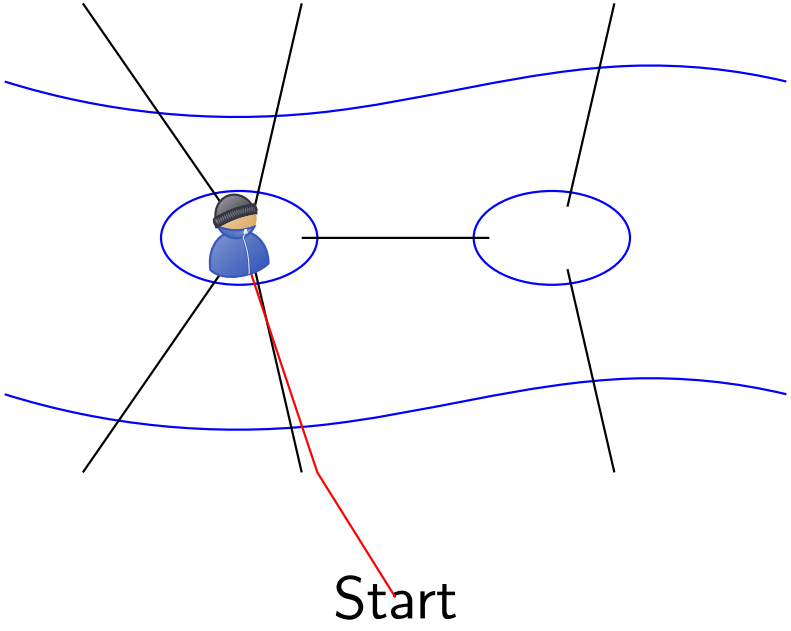

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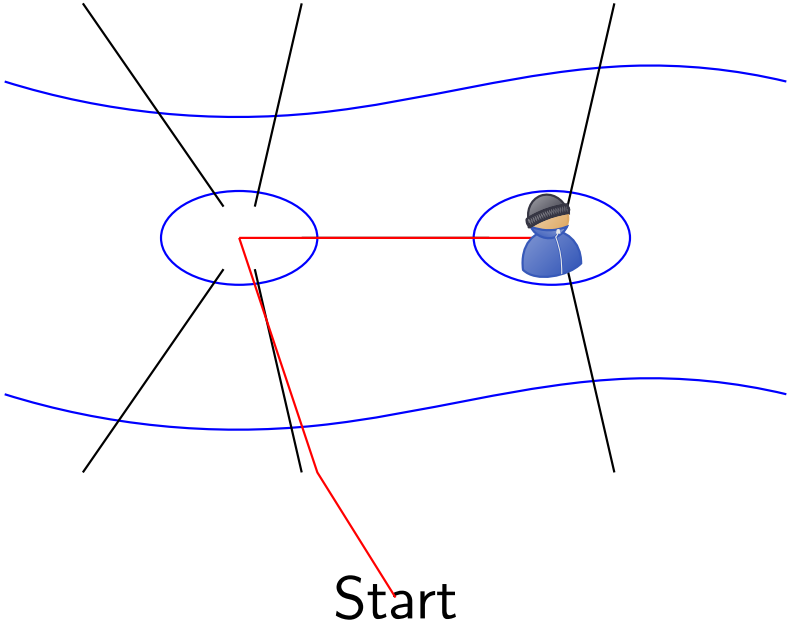

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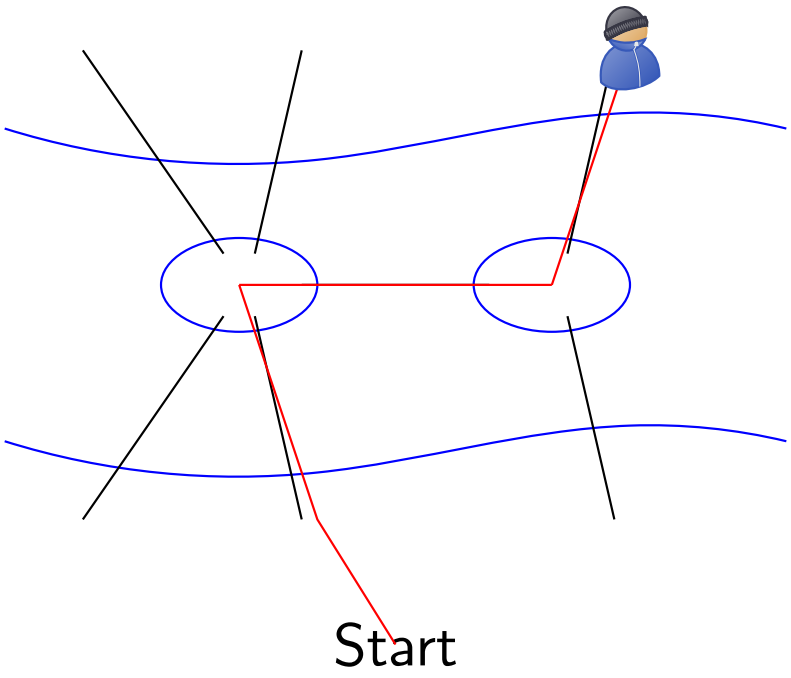

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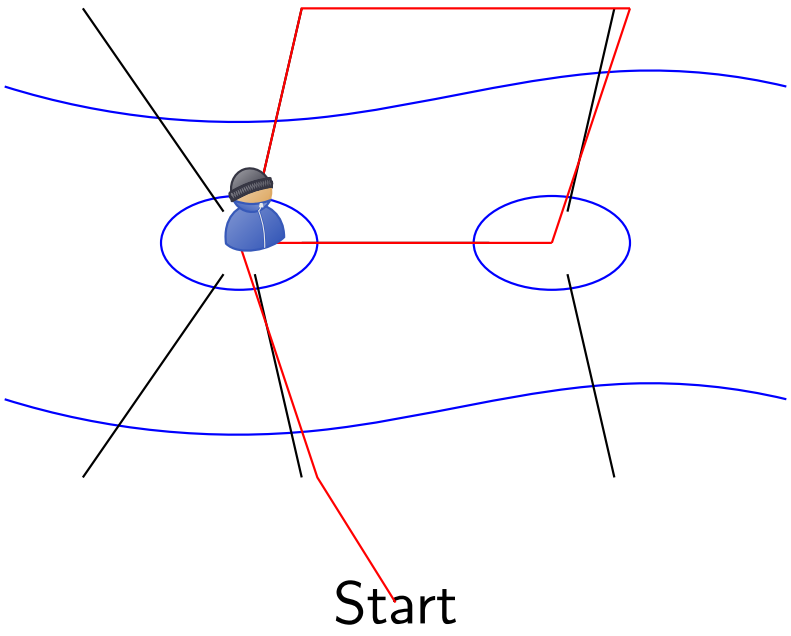

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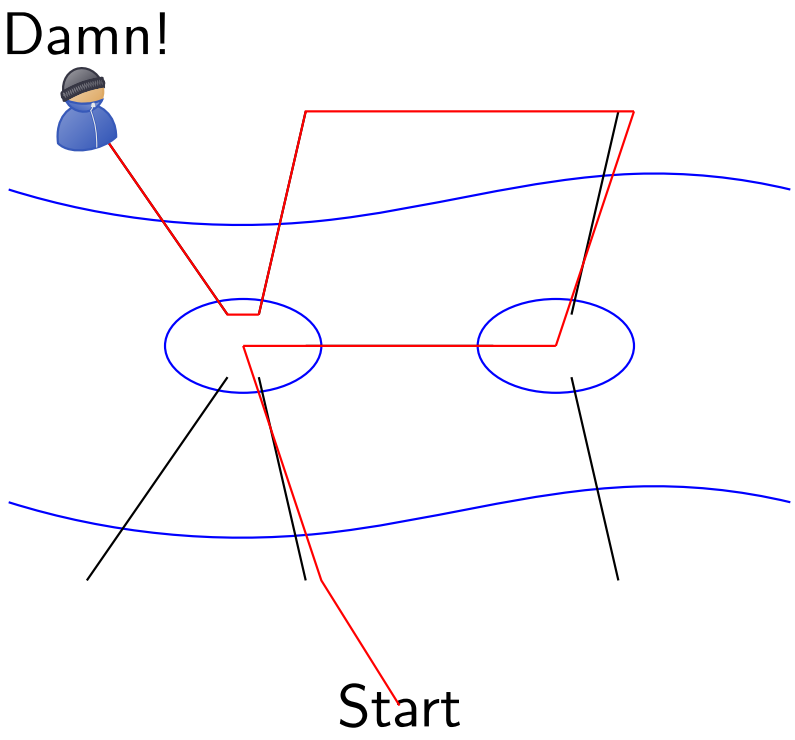

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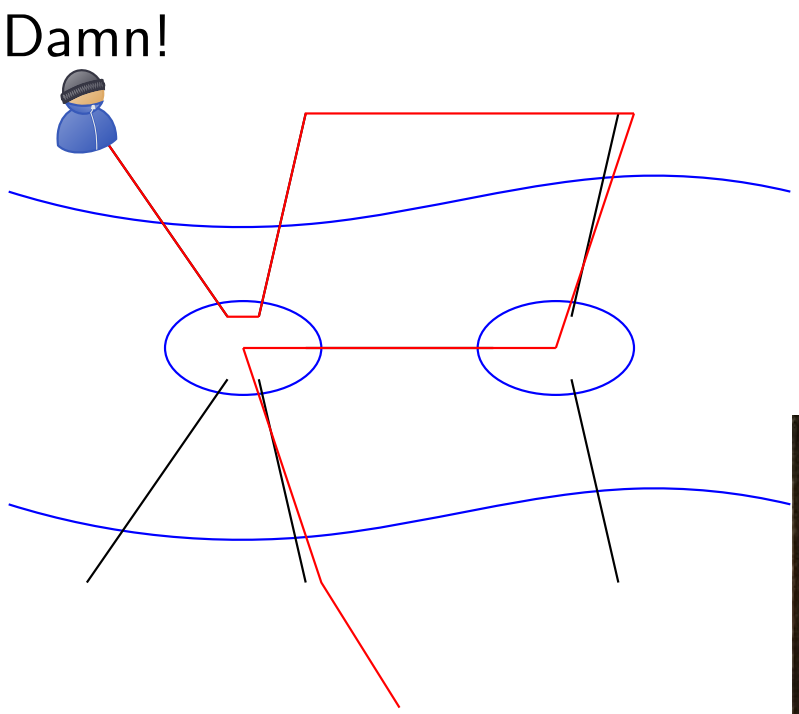
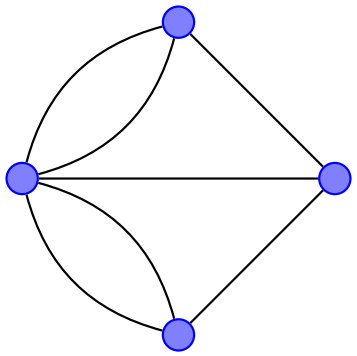


You aren't going to be able to do that!

The bridges of Konigsburg



Euler's Theorem

An eulerian circuit in a graph is a walk that visits every vertex, and every edge exactly once, and ends where it starts.
A graph is eulerian if it has an eulerian circuit.

Theorem

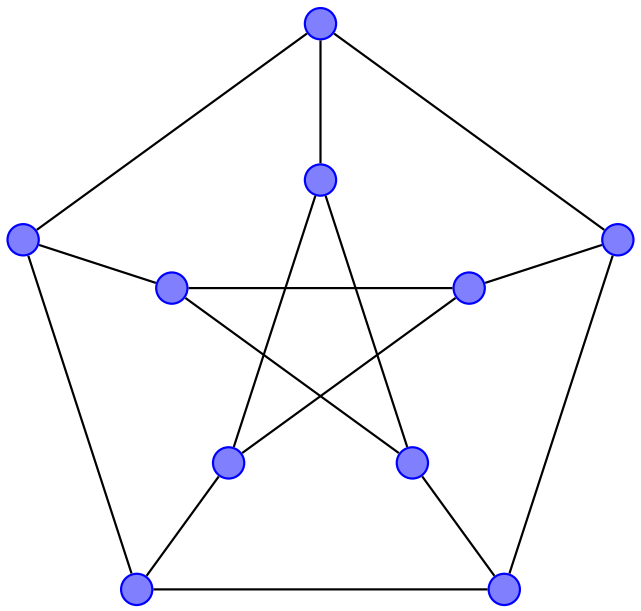
A graph is eulerian if and only if it is connected and every vertex has even degree.

Hamilton wants in on the bridge game action

What if we just want to visit every vertex?

Hamilton Cycles

A hamiton cycle in a graph is a cycle that visits every vertex.
A graph is hamiltonian if it has a hamilton cycle.



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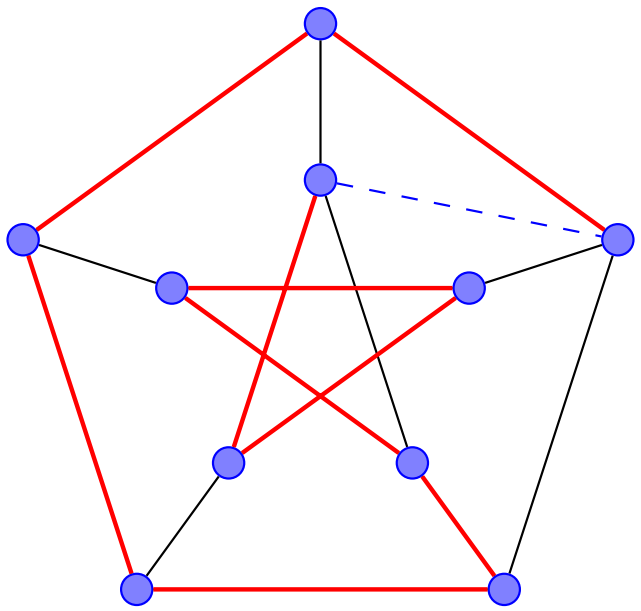


Hamilton cycles are trickier

Theorem

It is hard to decide if a graph has a hamilton cycle.
Indeed, NP-hard!

What should we do if a problem is hard

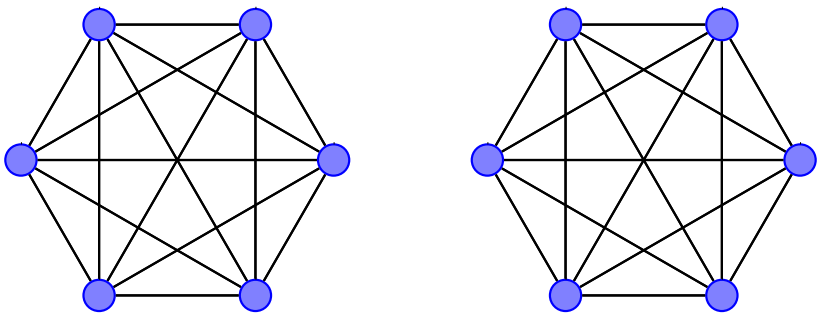
- Give up.
- Find conditions that make the problem easier.
- Approximate the problem.

Hamilton's Theorem

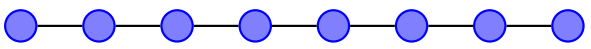
Theorem

If every vertex of a graph G on n vertices has degree at least $n/2$ then G is hamiltonian.

We need degree at least $n/2$

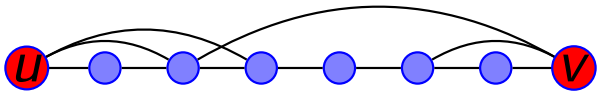


Why this is enough



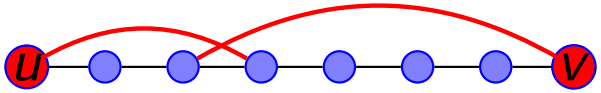
Take a longest path in G.

Why this is enough



All neighbours of the endpoints must be on the path, or there is a longer path.

Why this is enough



If u has neighbour i and v has neighbour $i - 1$...

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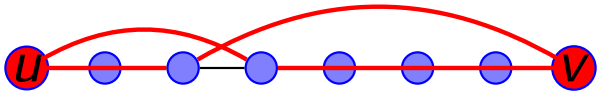
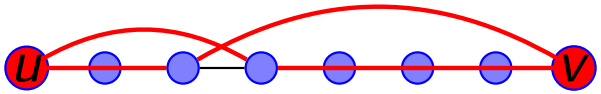


If u has neighbour i and v has neighbour $i-1$... then there is a hamilton cycle.

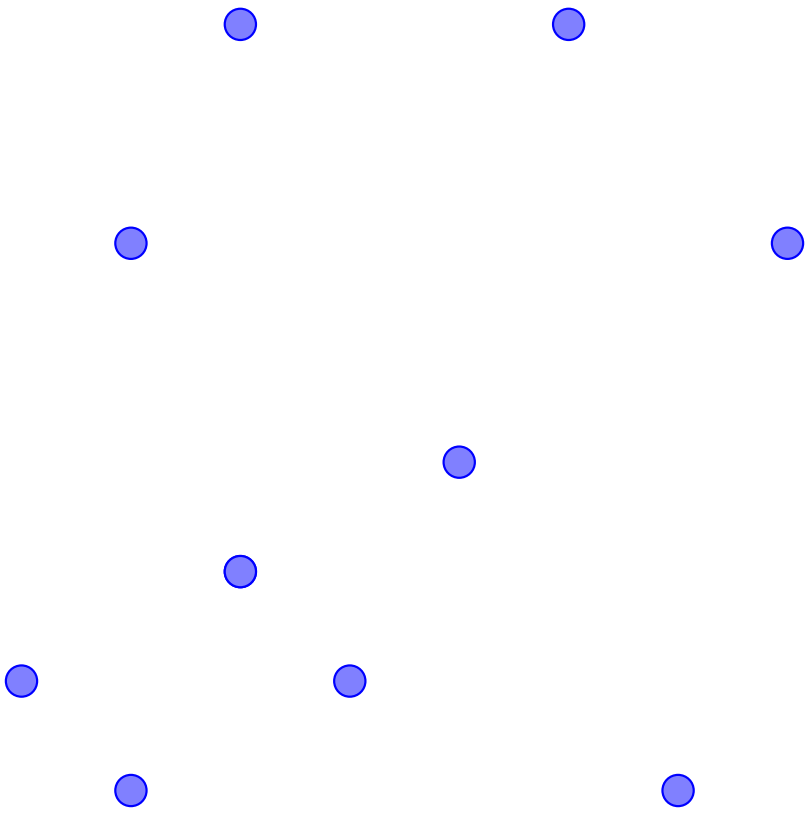
Why this is enough



So if u has $n/2$ neighbours then v has less then $n - n/2 = n/2$, or we have a hamilton cycle.

Graph Colourings

Radio towers and frequencies



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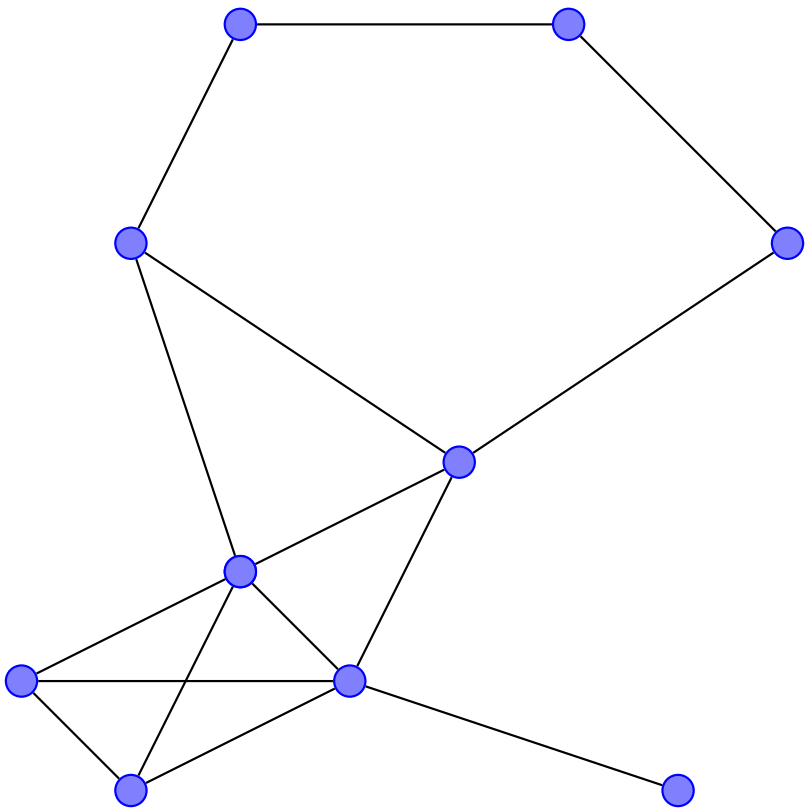

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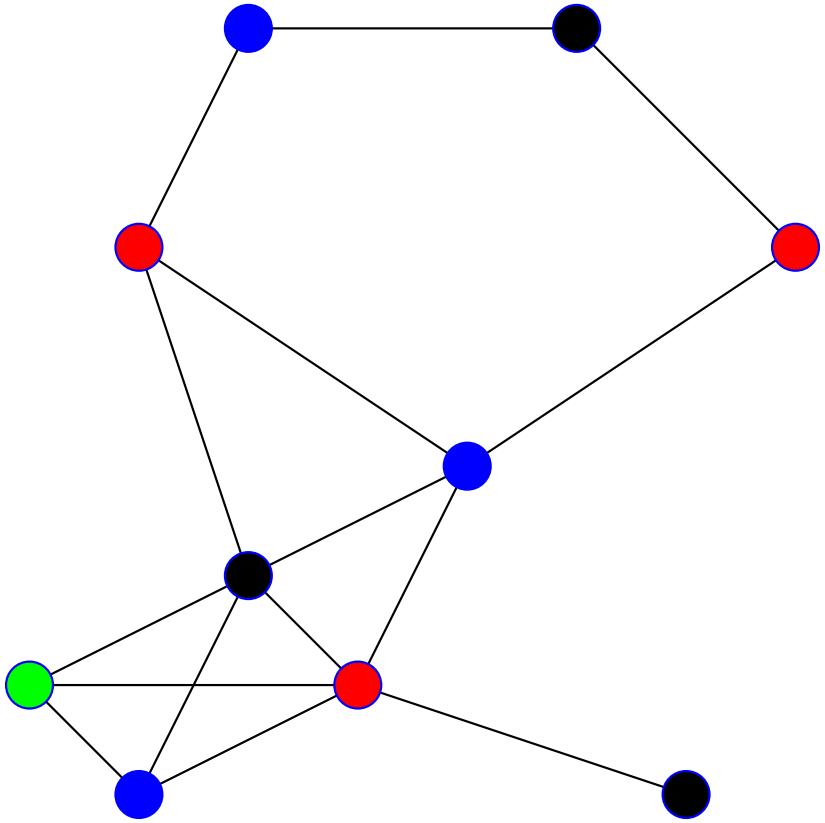
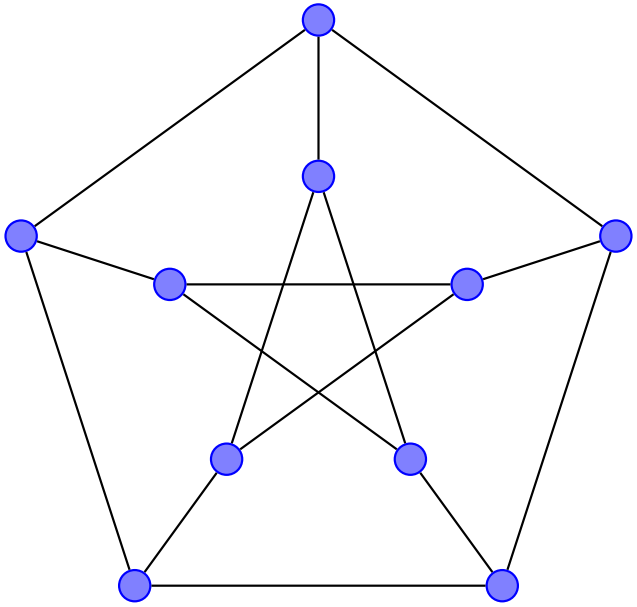


How many colours do we need for the Petersen graph?



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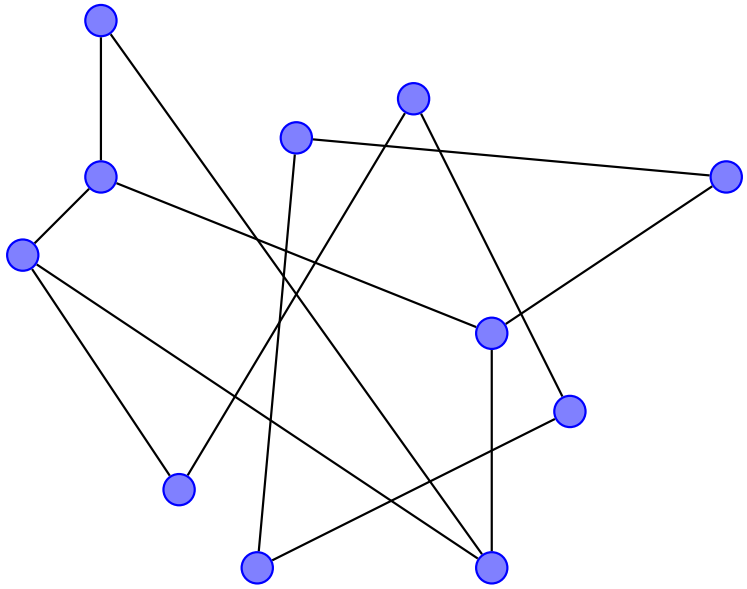
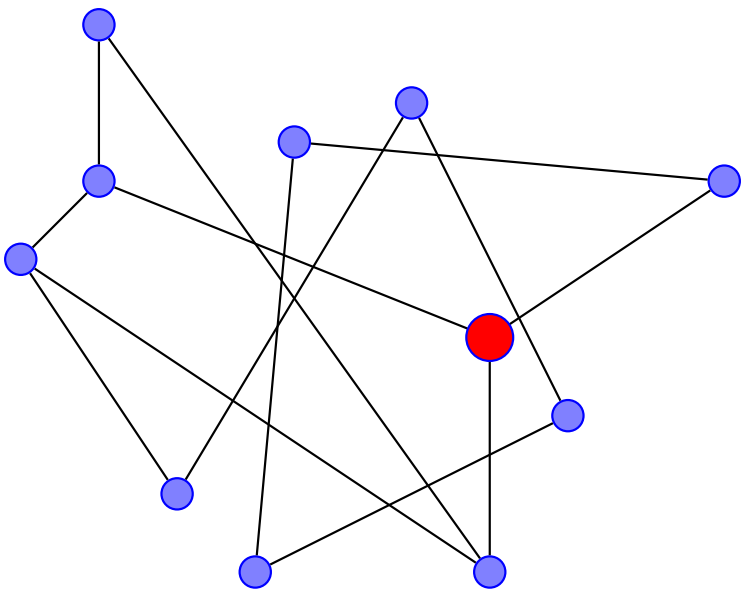


Deciding if a graph has a 2-colouring is easy.

How hard is colouring?



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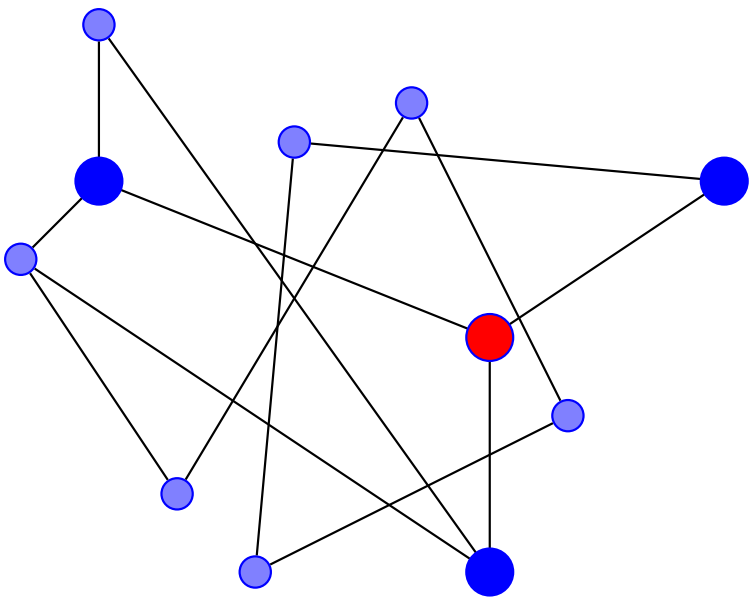

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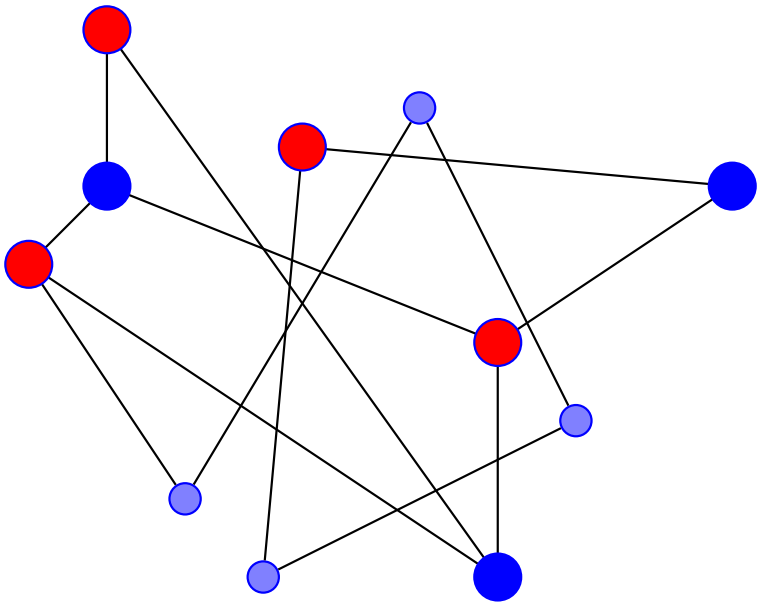

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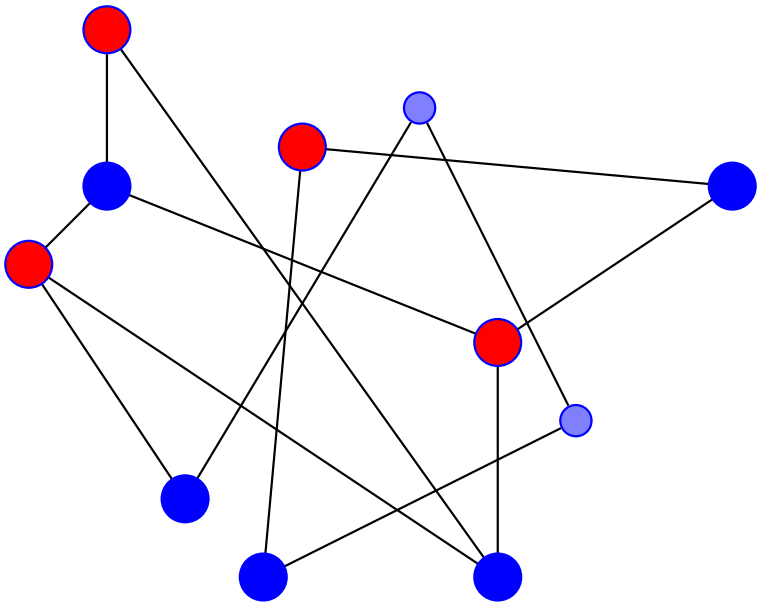

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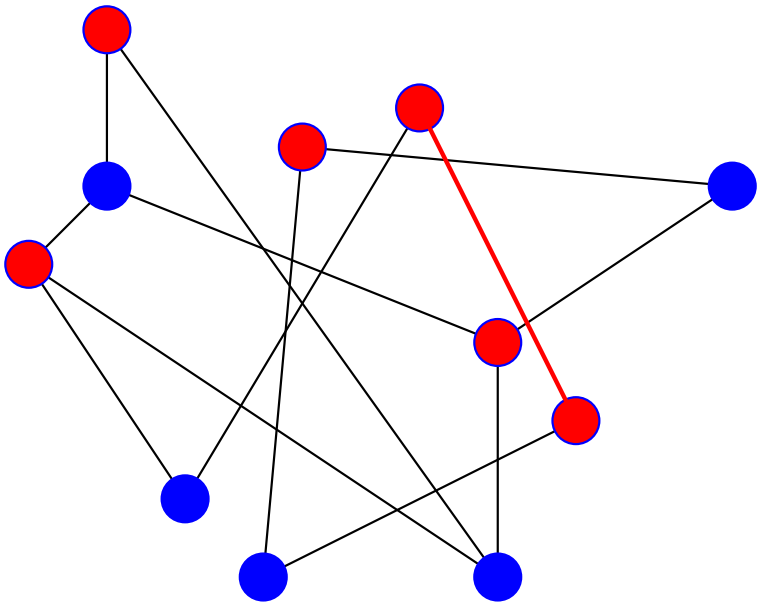

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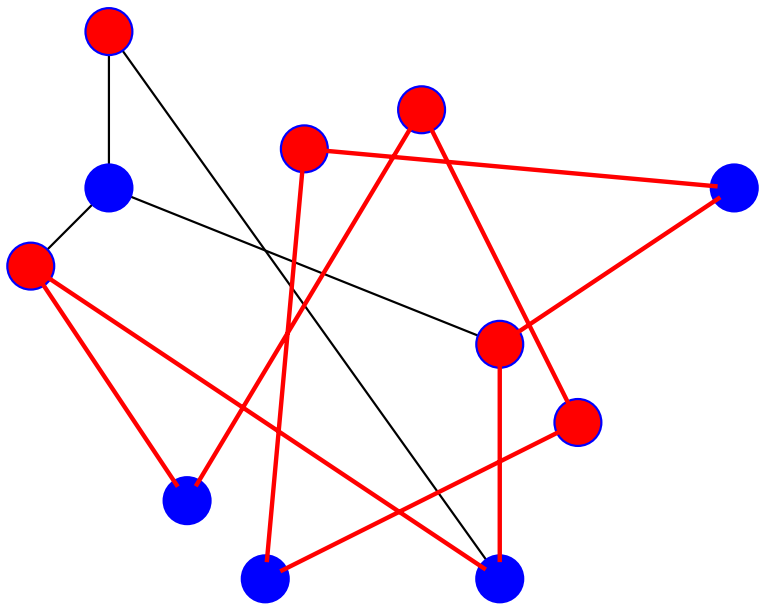
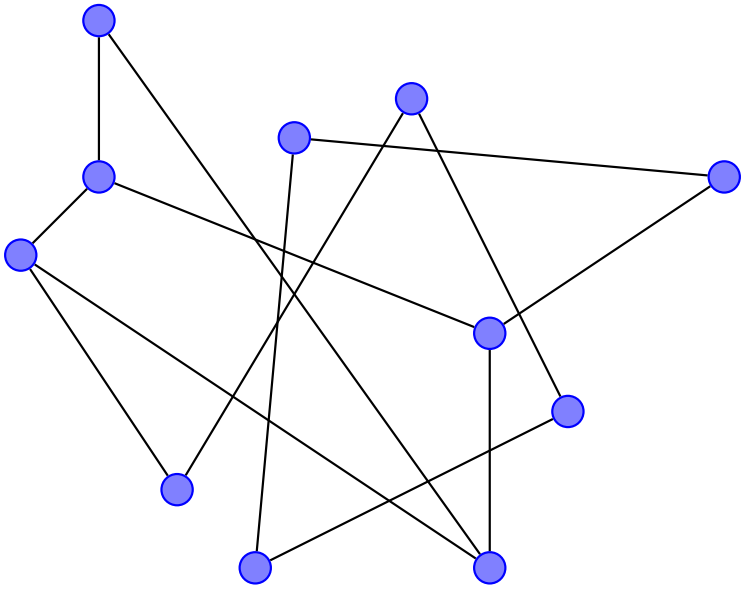


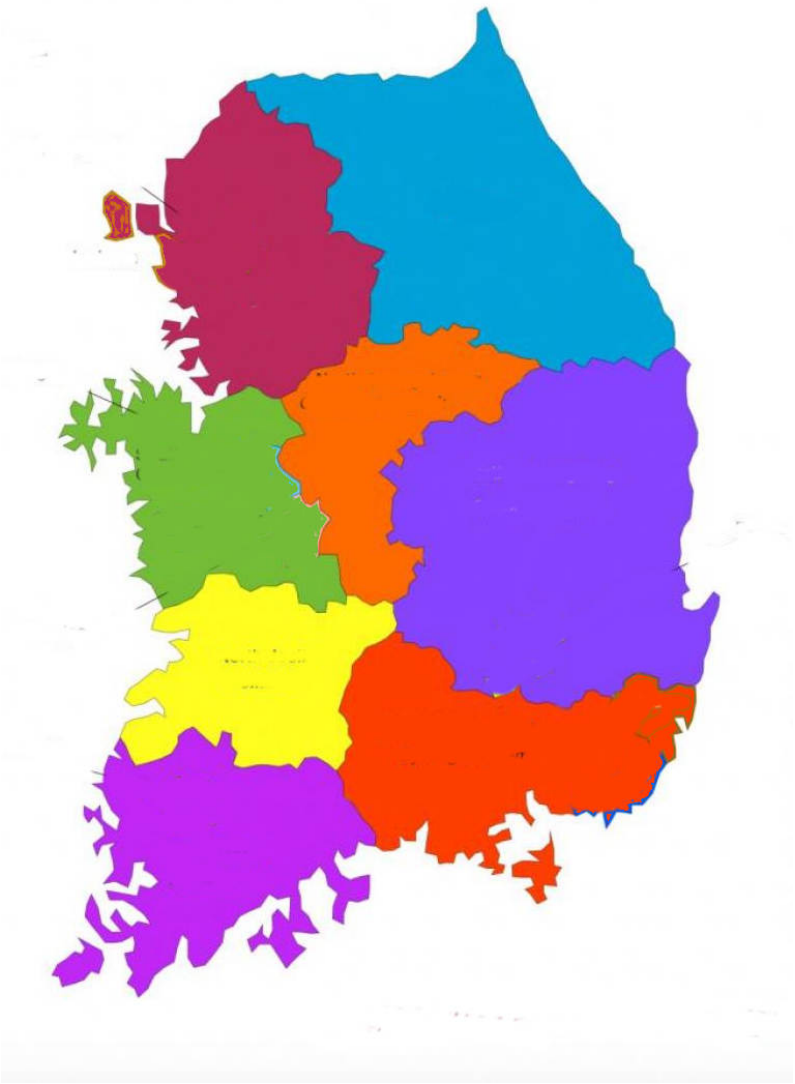
G has a 2-colouring if and only if it has no odd cycles.

How hard is colouring?



Deciding if a graph has a 3 or 4 or $k \geq 3$ colouring is NP-complete.

Colouring maps



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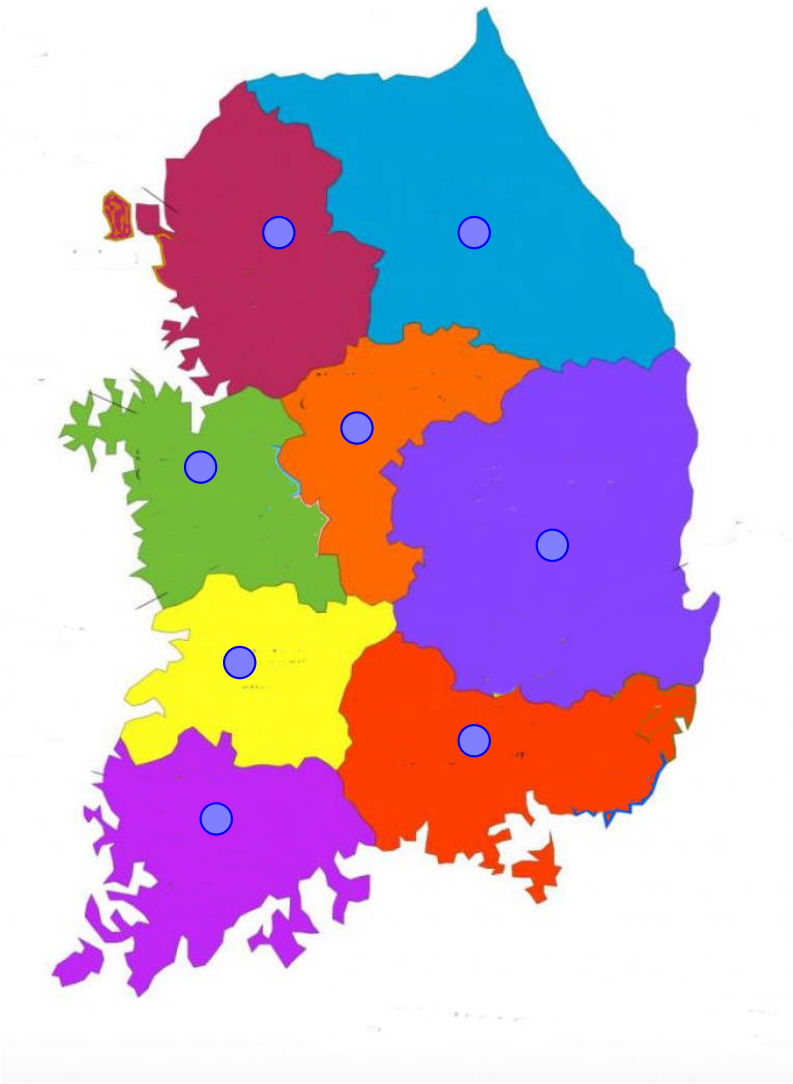

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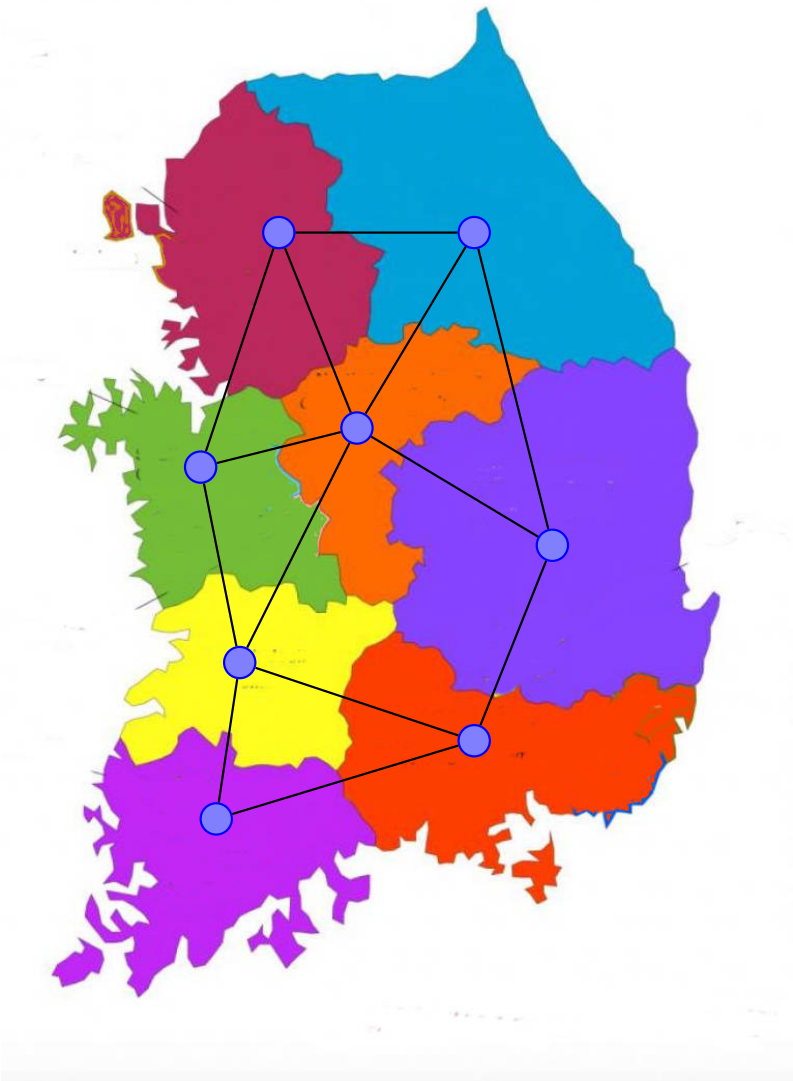

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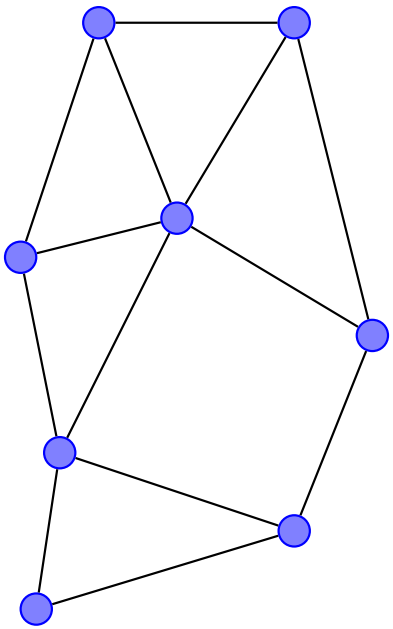


The four-colour theorem

Theorem (Haken, Appel)

Any graph that can be drawn in the plane without crossing edges can be coloured with 4 colours.

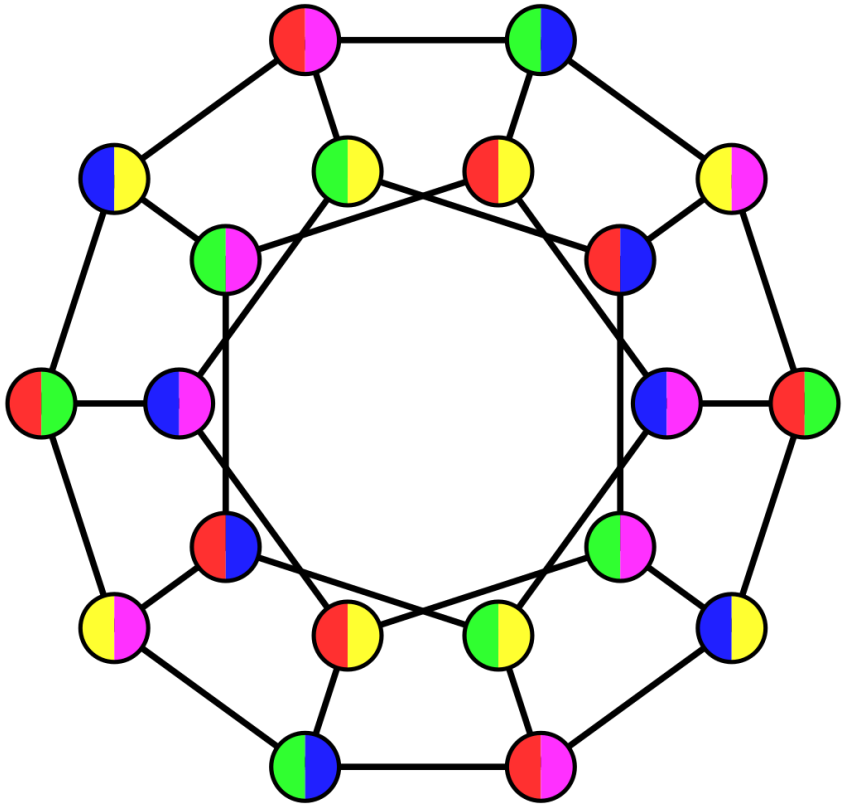
But...

Theorems

But: it's hard to decide if a given planar graph is 3-colourable.
But: any triangle-free planar graph is 3-colourable.
But ...

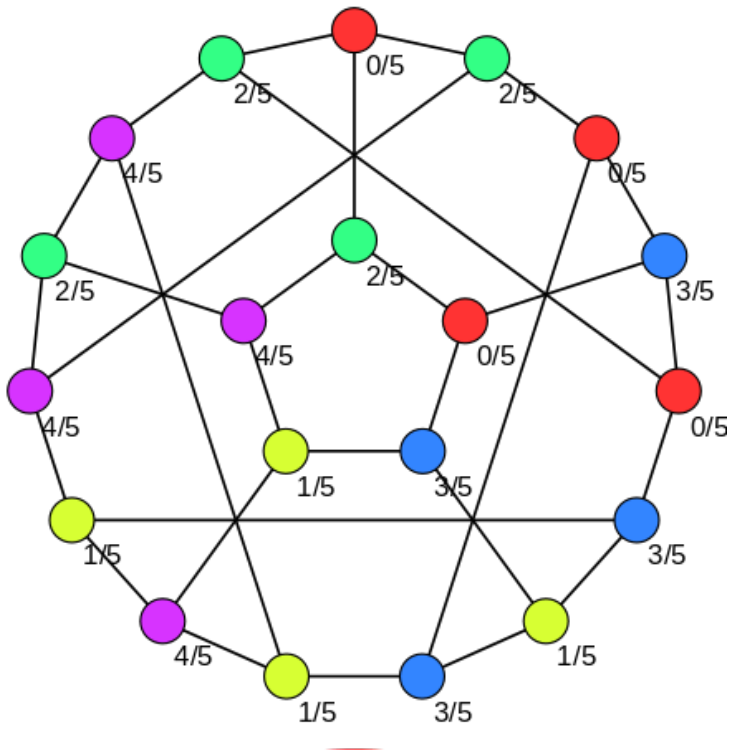
Other kinds of colourings

Fractional colourings:



Other kinds of colourings

Circular colourings:



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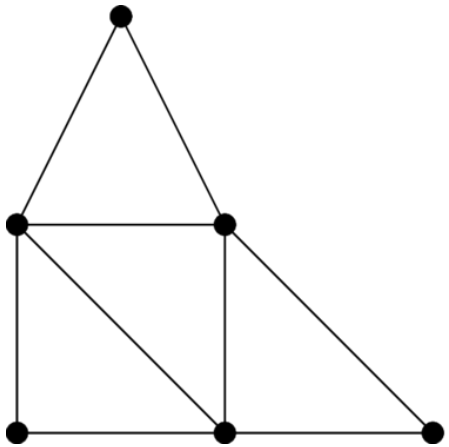
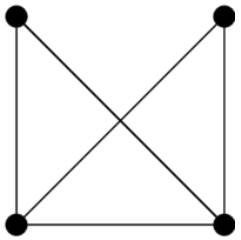
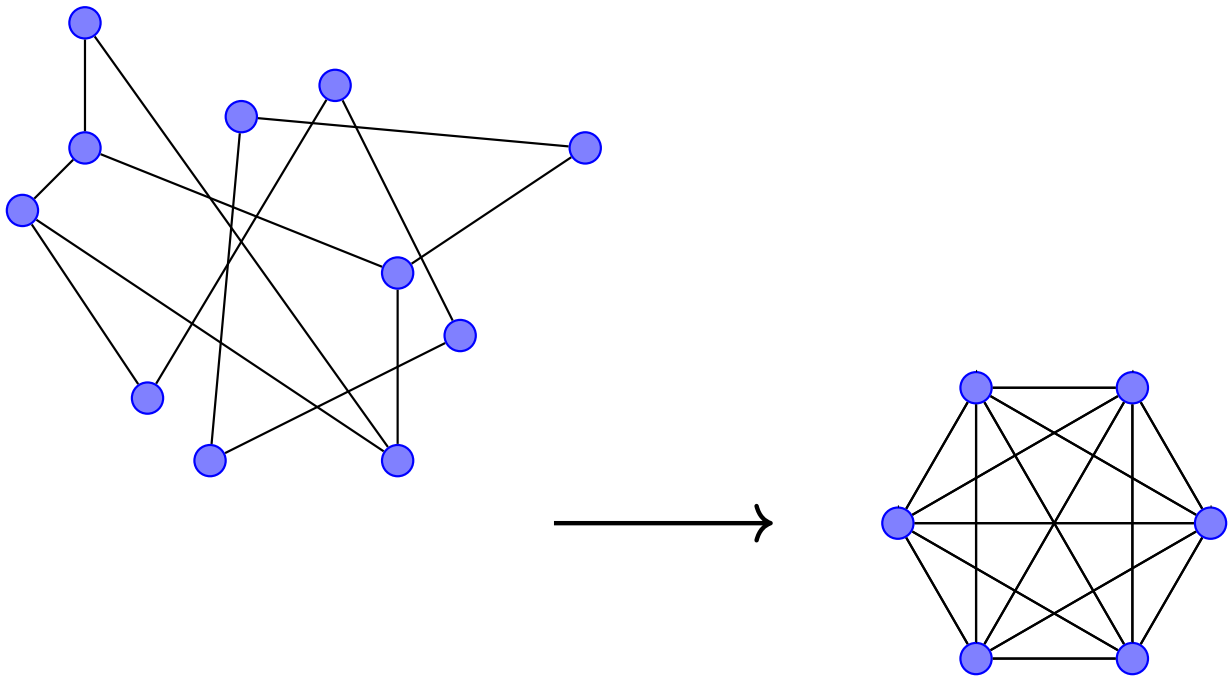




A homomorphism (or H-colouring) $G \to H$ of G is an edge preserving vertex map from G to H.

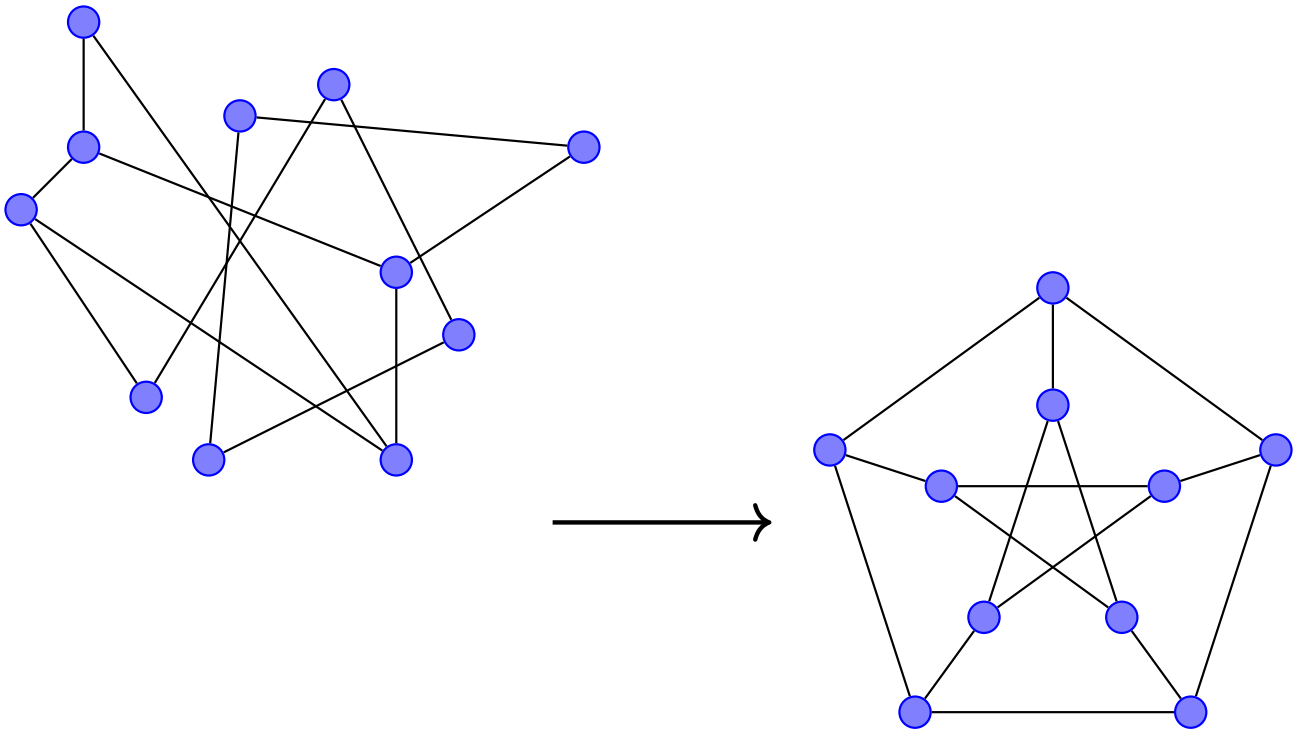
Graph Homomorphisms

k-colouring is homomorphism to a k-clique



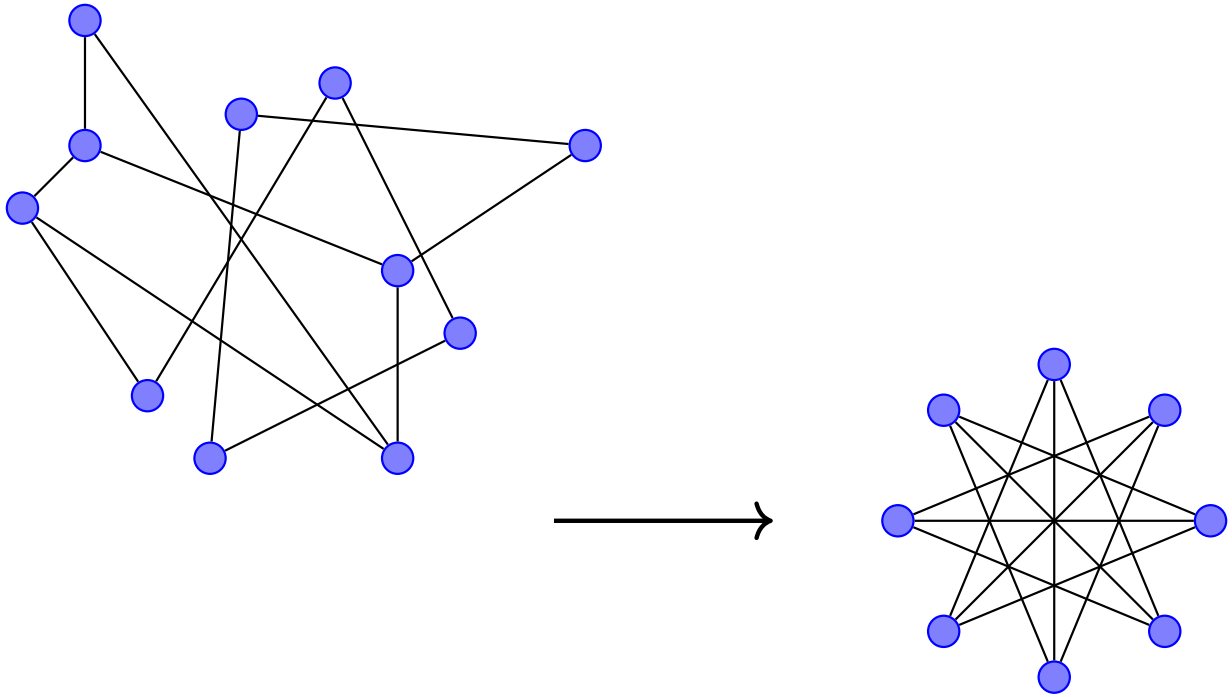
Graph Homomorphisms

fractional colouring is a homomorphism to a Knesser graph



Graph Homomorphisms

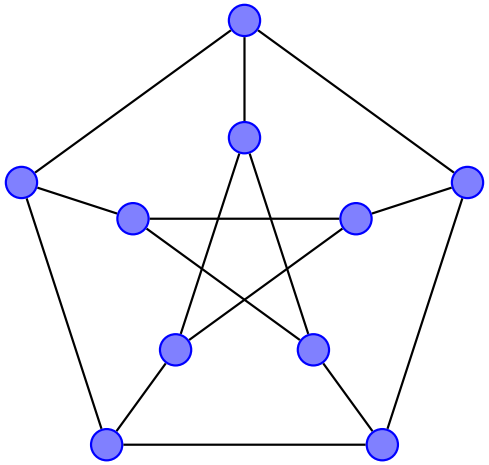
circular-colouring is homomorphsim to a circulant



But...

Theorems

But: it's hard to decide if a given planar graph is 3-colourable.
But: any triangle-free planar graph is 3-colourable.
But: it's hard to decide if a triangle-free planar graph maps to...

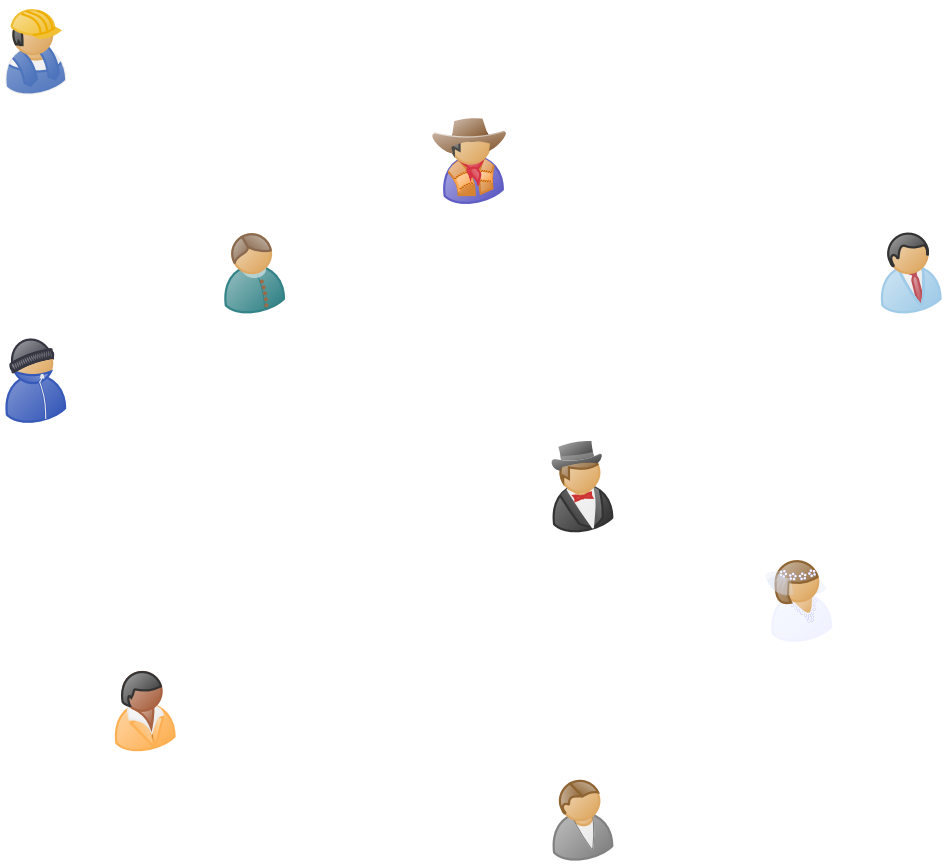


Ramsey Theory: a way different kind of colouring

Ramsey was at a party one day...

What are the chances?

Ramsey Numbers



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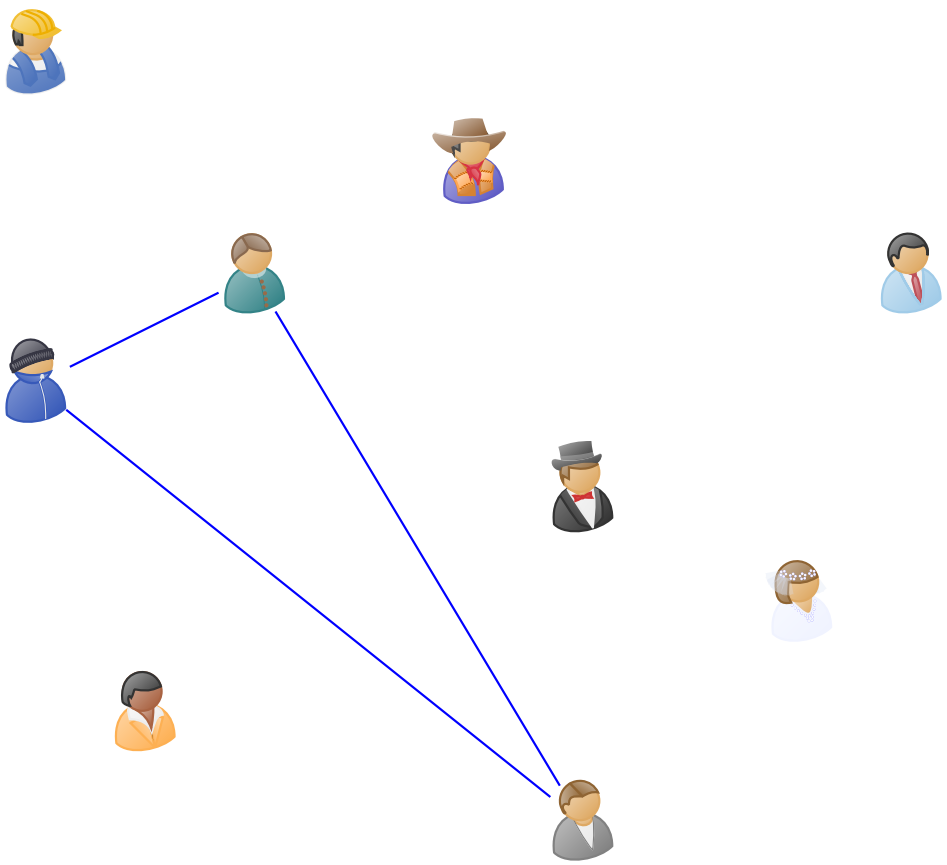

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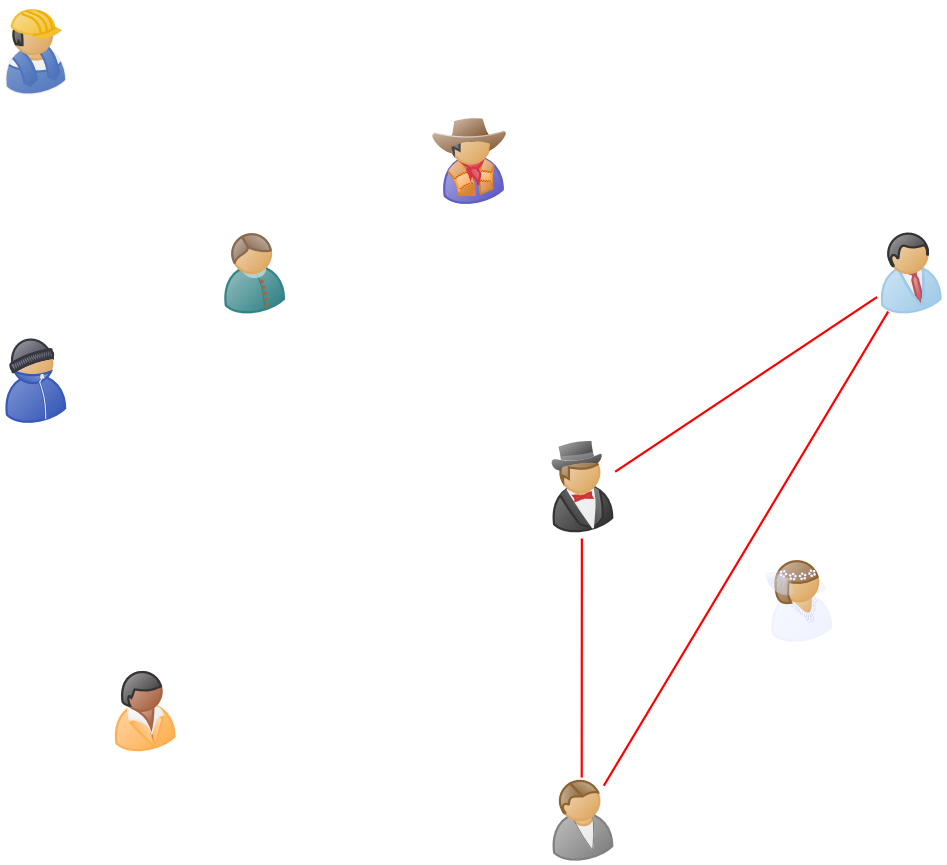

Ramsey Numbers

The ramsey number $R(m, n)$ is the minimum number r such that any blue-red edge colouring of K_r has a blue K_m or a red K_n.

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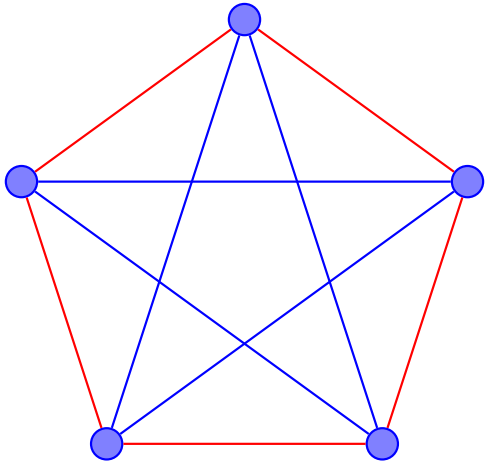
What is $R(3, 3)$?

$R(3,3) \geq 5$



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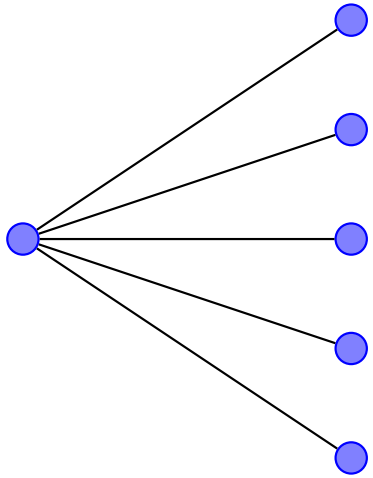

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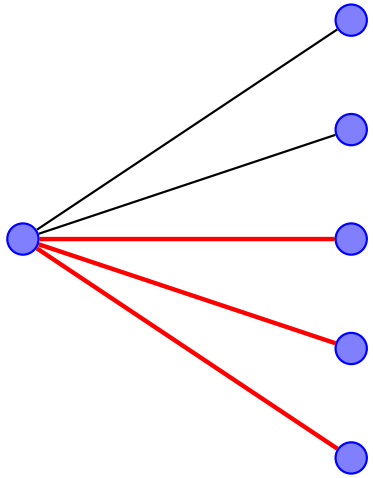

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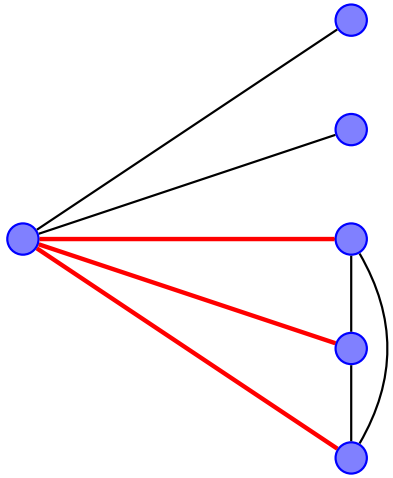

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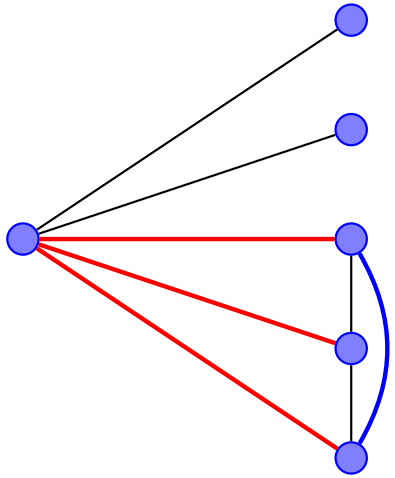

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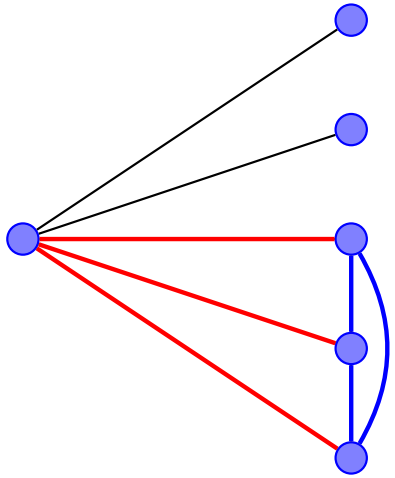

Ramsey Numbers

b \ r	3	4	5	6	7	8	9	10
3	6	9	14	18	23	28	36	40-43
4		18	25	35-41	49-61	56-84	73-115	92-149
5			43-49					
6				102-165				
7					205-540			

Problem 1:

A graph is k-regular if every vertex has degree k.

Number of vertices in a 3-regular graph

What is the least possible number of vertices in a 3-regular graph?

Problem 2:

The girth of a graph is the length of its shortest cycle.

Regular and girth

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... in a 3-regular graph of girth 5?

Problem 3:

The distance between two vertices is the length of the shortest path between them.

The diameter of a graph, is the minimum, over all pairs of vertices, of the distance between them:

$$\mathrm{diam}(G) = \max_{u,v \in V(G)} d(u, v).$$

Diameter and Max degree

What is the maximum number of vertices in a graph of diameter 4 having maximum degree 5?

Problem 4:

Mycielski

Find a 3-chromatic graph of girth 5.

...a 4-chromatic graph of odd girth 5.

Problem 5:

Let $G_{7,3}$ be the graph whose vertices are the three element subsets of the set $\{1, 2, \ldots, 7\}$ and in which the vertices U and V are adjacent if $|U \cap V| = 0$.

Kneser

Show that $G_{7,3}$ has chromatic number 3.

Hint:

Problem 5:

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Hint: Find an odd cycle to show chromatic number is at least 3.

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Hint: $f(S) = \min(S)$ is a 5 colouring. Why? Can you improve this?

Problem 6:

A graph is planar if it can be drawn in the plane without any crossing edges.

Petersen Planar

Is the Petersen Graph Planar?

Problem 7:

A P_2 is a path with 2 edges.

A decomposition of a graph G is a set of edge-disjoint subgraphs $H_1, \ldots, H_r$ whose union is G.

2-Path Decomposition

Show that any 4-regular graph G has a decomposition into copies of P_2.

Problem 8:

Chromatic number of the plane

Consider the graph U whose vertices are all points in $\mathbb{R}^2$. Points x and y are adjacent if $|x - y| = 1$. What is the chromatic number of U?

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